EXP 8

```
clc;
clear;
close all;

% Power Spectral Density (PSD) Analysis of PM Signals
% Using MATLAB

%% 1. System Parameters
Fs = 100000;            % Sampling frequency (Hz)
T = 0.1;                % Simulation time (seconds)
t = 0:1/Fs:T-1/Fs;      % Time vector

Am = 1;                 % Message amplitude
fm = 200;               % Message frequency (Hz)
Ac = 1;                 % Carrier amplitude
fc = 5000;              % Carrier frequency (Hz)
kp = 2;                 % Phase sensitivity (rad/V)

%% 2. Generate Message Signal
m = Am*cos(2*pi*fm*t);

%% 3. Generate Carrier Signal
carrier = Ac*cos(2*pi*fc*t);

%% 4. Generate Phase Modulated Signal
sPM = Ac*cos(2*pi*fc*t + kp*m);

%% 5. Calculate Power Spectral Density
% Welch's method
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```
[PSD, f] = pwelch(sPM, hamming(4096), 2048, 8192, Fs);

%% 6. Calculate PM Modulation Index
beta_PM = kp*Am;

%% 7. Plot Time-Domain Signals
figure('Name','PM Signal Analysis');

subplot(3,1,1);
plot(t,m,'b','LineWidth',1.2);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t,sPM,'r');
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% 8. Plot PSD in dB/Hz
subplot(3,1,3);
plot(f/1000,10*log10(PSD + eps),'k','LineWidth',1.2);
title('Power Spectral Density of PM Signal');
xlabel('Frequency (kHz)');
ylabel('PSD (dB/Hz)');
xlim([0 12]);
grid on;
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%% 9. Display Parameters
fprintf('Message Frequency = %.2f Hz\n',fm);
fprintf('Carrier Frequency = %.2f Hz\n',fc);
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('PM Modulation Index = %.2f radians\n',beta_PM);

disp('PM signal PSD analysis completed successfully.');
```

